

Estimating the sum of a series:

If $f(x)$ is a function such that $f(k) = a_k$, where f is continuous, decreasing, and $\sum a_n$ is convergent. So.

$$\int_{n+1}^{\infty} f(x) dx \leq R_n \leq \int_n^{\infty} f(x) dx$$

where $R_n = S - S_n$; $|R_n| = |S - S_n|$ is called the error.

For $\sum_{n=1}^{\infty} \frac{1}{n^3}$ and $n=10$ we have.

$$1.201664 \leq S \leq 1.202532.$$

If we approximate the sum S by the midpoint of this interval, then the error is at most half the length of the interval. So.

$$\sum_{n=1}^{\infty} \frac{1}{n^3} \approx 1.2021 \quad \text{with error} < 0.005.$$

11.4 Comparison test.

The Direct Comparison test.

Let's consider $\sum a_n$ and $\sum b_n$ are series with positive terms.

- i). If $\sum b_n$ is convergent and $a_n \leq b_n$ then $\sum a_n$ is convergent.
- ii). If $\sum b_n$ is divergent and $b_n \leq a_n$ then $\sum a_n$ is divergent.

Example: Determine whether the series $\sum_{n=1}^{\infty} \frac{5}{2n^2 + 4n + 3}$ converges or diverges.

Sol: $b^2 - 4ac = 16 - 4(2)(3) < 0$. Notice that $\frac{5}{2n^2 + 4n + 3} > 0$ for all n .

Let consider $\frac{5}{2n^2 + 4n + 3} \leq \frac{5}{2n^2}$.

and $\sum_{n=1}^{\infty} \frac{5}{2n^2} = \frac{5}{2} \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$ b/c is a p -series with $p=2 > 1$.

Now, by the comparison thm $\sum_{n=1}^{\infty} \frac{5}{2n^2+4n+3}$ is convergent.

Example. Test the series $\sum_{k=1}^{\infty} \frac{\ln k}{k}$ for convergence or divergence.

Sln. let consider $\frac{\ln n}{n} \geq \frac{1}{n}$ and since $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent.

So by the comparison thm $\sum_{n=1}^{\infty} \frac{\ln n}{n}$ is divergent.

Limit Comparison test.

Thm. Suppose $\sum a_n$ and $\sum b_n$ are series with positive terms.

If

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c.$$

where c is finite number $c > 0$, then either both series converges or both diverge.

Rmk. Make sure that you have in the denominator the general term of the series $\sum b_n$ and we know if $\sum b_n$ is convergence or divergent series.

Example. test the series $\sum_{n=1}^{\infty} \frac{1}{2^{n-1}}$ for convergence or divergence.

Sln. Let consider $b_n = \frac{1}{2^n} = \left(\frac{1}{2}\right)^n$ and $\sum_{n=1}^{\infty} \frac{1}{2^n} < \infty$ b/c is a geometric series with $a=1$, $r=\frac{1}{2}$ and $|r| < 1$.

let $a_n = \frac{1}{2^{n-1}}$ and

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{\frac{1}{2^{n-1}}}{\frac{1}{2^n}} = \lim_{n \rightarrow \infty} \frac{2^n}{2^{n-1}} = \lim_{n \rightarrow \infty} \frac{1}{1 - \frac{1}{2}} = 1$$

so, $1 > 0$ and by the limit comparison test since $\sum b_n$ is convergent then $\sum_{n=1}^{\infty} \frac{1}{2^{n-1}}$ is convergent.

Example: Determine whether the series $\sum_{n=1}^{\infty} \frac{2n^2+3n}{\sqrt{5+n^5}}$ converges or diverges.

Soln: Let $b_n = \frac{2n^2}{\sqrt{n^5}} = \frac{2}{\sqrt{n}} = \frac{2}{n^{1/2}}$ and $\sum_{n=1}^{\infty} \frac{2}{n^{1/2}}$ is divergent

b/c is a p-series with $p = 1/2 < 1$. Now,

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{a_n}{b_n} &= \lim_{n \rightarrow \infty} \frac{\frac{2n^2+3n}{\sqrt{5+n^5}}}{\frac{2}{\sqrt{n}}} = \lim_{n \rightarrow \infty} \frac{(2n^2+3n)\sqrt{n}}{2\sqrt{5+n^5}} \\ &= \frac{1}{2} \lim_{n \rightarrow \infty} \frac{2n^{5/2} + 3n^{3/2}}{\sqrt{5+n^5}} \\ &= \frac{1}{2} \lim_{n \rightarrow \infty} \frac{2n^{5/2}/n^{5/2} + 3n^{3/2}/n^{5/2}}{\sqrt{5+n^5}} \\ &= \frac{1}{2} \lim_{n \rightarrow \infty} \frac{2 + 3/n}{\sqrt{5/n^5 + 1}} = \frac{1}{2} \left(\frac{2+0}{\sqrt{0+1}} \right) = 1 > 0 \end{aligned}$$

By the limit comparison thm since $\sum b_n$ is divergent so

$$\sum_{n=1}^{\infty} \frac{2n^2+3n}{\sqrt{5+n^5}} \text{ is divergent.}$$

Example: $\sum_{n=1}^{\infty} \frac{n^2-3n}{\sqrt{n^5+2}}$. $a_n = \frac{n^2-3n}{\sqrt{n^5+2}}$ $b_n = \frac{n^2}{\sqrt{n^5}} = \frac{n^2}{n^{5/2}} = \frac{1}{\sqrt{n}}$
and $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} = \sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$ is divergent by p-series with $p = 1/2 < 1$.

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{a_n}{b_n} &= \lim_{n \rightarrow \infty} \frac{\sqrt{n}(n^2-3n)}{\sqrt{n^5+2}} = \lim_{n \rightarrow \infty} \frac{n^{5/2}-3n^{3/2}}{\sqrt{n^5+2}} \\ &= \lim_{n \rightarrow \infty} \left(\frac{1-3/n}{\sqrt{1+2/n^5}} \right) = 1 > 0 \text{ By the limit comparison} \end{aligned}$$

test $\sum_{n=1}^{\infty} \frac{n^2 - 3n}{\sqrt{n^5 + 2}}$ is divergent.

Problem: let consider $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^3}$ is alternating series

If $|a_n| \rightarrow 0$ as $n \rightarrow \infty$ and $|a_n|$ is decreasing. then $\sum_{n=1}^{\infty} a_n$ is convergent.

let consider $a_n = \frac{(-1)^n}{n^3}$ then $|a_n| = \frac{1}{n^3} \rightarrow 0$ as $n \rightarrow \infty$.

Notice that $(n+1) \geq n \Rightarrow (n+1)^3 \geq n^3 \Rightarrow \frac{1}{n^3} \geq \frac{1}{(n+1)^3}$

$|a_n| \geq |a_{n+1}|$ for all $n \geq 1$

So is decreasing. Then $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^3}$ is convergent by alternating test.

We can try approximate using.

$$|\text{error}| = |R_n| = |S - S_n|$$

Find the smallest n such that $|\text{error}| < \frac{1}{100}$

$$|S - S_n| \leq |a_{n+1}| < \frac{1}{100}$$

$$\frac{1}{(n+1)^3} < \frac{1}{100}$$

$$\Rightarrow (n+1)^3 > 100$$

when $n=4$ we have $(4+1)^3 > 100 \checkmark$

$n=3$ we have $(3+1)^3 > 100 \times$.

So

$$n=4$$